
Table of Contents

hw3_me501_Q7.m	1
Continuous time matrices for consideration	1
Checking continuous time eigenvalues	1
Exact discretization and eigenvalues	2

hw3_me501_Q7.m

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```
close all;
clc;

disp('Checking stability of exact discretizations.')

Checking stability of exact discretizations.
```

Continuous time matrices for consideration

```
A1 = [0 1; -20 0];

A2 = [0 1; -20 -9];

g = 10;
m = 1;
c = 1;

A3 = [0 0 0 1 0 0;
      0 0 0 0 1 0;
      0 0 0 0 0 1;
      0 0 -g -c/m 0 0;
      0 0 0 0 -c/m 0;
      0 0 0 0 0 0];
```

Checking continuous time eigenvalues

```
[V1, D1] = eig(A1);
[V2, D2] = eig(A2);
[V3, D3] = eig(A3);

% Pull out the diagonal elements for ease of viewing:
disp('Diagonal elements of D for A1, A2, then A3:')
diag(D1)
diag(D2)
diag(D3)

% To calculate if we get enough eigenvectors to pass the multiplicity test,
% an easy way is to check the rank of the V that matlab produces. This
% tells us how many linearly independent columns of the matrix there are.
```

```
disp('Rank of V for A3:')
rank(V3)
```

Diagonal elements of D for A1, A2, then A3:

```
ans =
```

```
0.0000 + 4.4721i
0.0000 - 4.4721i
```

```
ans =
```

```
-4.0000
-5.0000
```

```
ans =
```

```
0
0
-1
0
-1
0
```

Rank of V for A3:

```
ans =
```

```
4
```

Exact discretization and eigenvalues

```
% We hypothesize that the sampling time shouldn't matter under the exact
% discretization - try it out and confirm for yourself!
```

```
dt = 0.01;
```

```
% Our claim is that  $x(k) = A_d^k x(0)$ , where  $A_d = e^{(A \cdot \text{deltaT})}$ 
```

```
disp('Exact discretization of A1, A2, and A3:')
```

```
A1d = expm(A1*dt)
```

```
A2d = expm(A2*dt)
```

```
A3d = expm(A3*dt)
```

```
% Check the eigenvalues of the discretized dynamics
```

```
[V1d, D1d] = eig(A1d);
```

```
[V2d, D2d] = eig(A2d);
```

```
[V3d, D3d] = eig(A3d);
```

```
% Pull out the diagonal elements for ease of viewing:
```

```
disp('Diagonal elements of D for A1d, A2d, then A3d:')
```

```

lambda1d = diag(D1d)
lambda2d = diag(D2d)
lambda3d = diag(D3d)

disp('Magnitude (modulus) of the discrete time eigenvalues:')
abs(lambda1d)
abs(lambda2d)
abs(lambda3d)

disp('Rank of V for A3d:')
rank(V3d)

```

Exact discretization of A1, A2, and A3:

A1d =

```

    0.9990    0.0100
   -0.1999    0.9990

```

A2d =

```

    0.9990    0.0096
   -0.1912    0.9130

```

A3d =

```

    1.0000         0   -0.0005    0.0100         0   -0.0000
         0    1.0000         0         0    0.0100         0
         0         0    1.0000         0         0    0.0100
         0         0   -0.0995    0.9900         0   -0.0005
         0         0         0         0    0.9900         0
         0         0         0         0         0    1.0000

```

Diagonal elements of D for A1d, A2d, then A3d:

lambda1d =

```

    0.9990 + 0.0447i
    0.9990 - 0.0447i

```

lambda2d =

```

    0.9608
    0.9512

```

lambda3d =

```

    1.0000
    1.0000
    0.9900

```

```
1.0000
0.9900
1.0000
```

Magnitude (modulus) of the discrete time eigenvalues:

```
ans =
```

```
1.0000
1.0000
```

```
ans =
```

```
0.9608
0.9512
```

```
ans =
```

```
1.0000
1.0000
0.9900
1.0000
0.9900
1.0000
```

Rank of V for A3d:

```
ans =
```

```
5
```

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